

PROVISIONAL PATENT SPECIFICATION

Title:

Systems and Methods for Integrated Admissibility-Governed Reasoning Architecture and Reduced Runtime Governance Deployment

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1. FIELD OF THE INVENTION

The invention relates to artificial intelligence, machine reasoning systems, structured semantic processing, epistemic governance, and runtime control of machine-generated outputs.

More particularly, the invention concerns systems and methods for integrating a structured reasoning-construction architecture with a commitment-control layer, and for deploying the commitment-control layer in reduced or hybrid form between an application, model interface, language generation system, downstream process, or other output destination.

The invention further concerns governance of whether a candidate output, inference, determination, recommendation, action, claim, or state transition may proceed to commitment, delivery, consequence, execution, or downstream reliance.

2. CROSS-REFERENCE TO RELATED APPLICATIONS

This application relates to, builds upon, and may claim priority from one or more of the following Australian provisional patent applications filed by the present inventor:

Application No. 2025905835, filed 27 November 2025, entitled "Aurora — Autonomous Conceptual Reasoning Architecture";

Application No. 2025905860, filed 28 November 2025, entitled "Systems and Methods for Implementing a Persistent Existence Frame (PEF) in Cognitive and Machine Reasoning Architectures";

Application No. 2025905885, filed 29 November 2025, entitled "Autonomous Compositional Reasoning Architecture Integrated with Persistent Existence Frame (Aurora + PEF)";

Application No. 2025906132, filed 8 December 2025, entitled "Systems and Methods for Explicit Compositional Reasoning in a Continuous-Present Cognitive Architecture";

Application No. 2025906680, filed 19 December 2025, entitled "Admissibility-Controlled Epistemic Governance for Machine Reasoning."

Each of the foregoing disclosures is incorporated herein by reference for all purposes permitted under applicable law.

The present disclosure does not replace or narrow the earlier disclosures. It provides an integration, deployment, and governance bridge among them, including reduced deployment of a commitment-control layer, hybrid deployment of reasoning and governance components, and explicit governance records for admissibility-controlled machine reasoning systems.

3. BACKGROUND OF THE INVENTION

Existing artificial intelligence systems frequently generate fluent outputs without determining whether the underlying inference, determination, recommendation, claim, or action is admissible.

Existing mitigation systems often operate as post-hoc guardrails, filters, confidence thresholds, moderation layers, retrieval augmentation, or policy checks surrounding a statistical model. Such approaches may restrict content or adjust generated text but do not necessarily govern whether a candidate determination is structurally permitted to become consequential.

A further problem arises when a complete reasoning architecture is reduced for deployment. A full architecture may include structured conceptual state, compositional primitives, persistent state substrate, parallel interpretation maintenance, and governed collapse logic. However, practical deployments may require a smaller runtime governance module interposed between a model interface and an application. Without a clearly defined relationship between the full architecture and the reduced deployment, a reduced product embodiment may be mistaken for the entire invention, thereby narrowing the architectural scope.

A further problem arises in operational environments where conditions change without generating explicit notification events. A previously valid authority, evidence basis, source credential, policy condition, supplier status, sanctions status, or institutional state may become stale, expired, superseded, or contradicted without any revocation signal being delivered to the governed system. In such circumstances, continued operation under the prior admissibility determination constitutes reliance on a justification that is no longer current. Systems that treat the absence of a revocation event as permission to continue fail to detect this class of silent admissibility decay.

There is therefore a need for:

(a) a unified architecture in which reasoning-construction and commitment-control are defined as interoperable layers;

(b) a reduced runtime governance embodiment that preserves the same admissibility invariant without requiring the full reasoning-construction stack;

(c) hybrid deployments in which the reduced governance module may receive state, constraints, evidence, authority, audit, or admissibility metadata from one or more upstream or parallel systems;

(d) governance of candidate outputs before delivery, commitment, consequence, execution, or downstream reliance;

(e) an admissibility-state indicator that may be exposed or enforced without constituting commitment;

(f) governance records sufficient to reconstruct, audit, replay, or evaluate admissibility determinations;

(g) claim language preserving the full architecture while allowing reduced deployment; and

(h) detection and governance of silent admissibility decay arising from stale, expired, superseded, or unvalidated external orientation conditions.

4. SUMMARY OF THE INVENTION

The invention provides an integrated machine reasoning and governance architecture comprising a reasoning-construction layer and a commitment-control layer.

The reasoning-construction layer may comprise structured conceptual state, concept kernels, reflective semantic topology, roles, domains, spans, compositional primitives, interpretability evaluation, parallel interpretation maintenance, Aurora Block processing, Persistent Existence Frame operation, or any functional equivalent thereof.

The commitment-control layer governs whether a candidate state, candidate interpretation, output, determination, recommendation, action, claim, inference, or

model-generated response may become committed, delivered, executed, relied upon, or otherwise consequential.

In full-stack embodiments, the commitment-control layer operates on structured conceptual state produced by the reasoning-construction layer.

In reduced embodiments, the commitment-control layer may be deployed independently as a runtime governance module interposed between an application and a model interface, language generation system, downstream system, automated process, user interface, or output destination.

The reduced runtime governance module does not require deployment of the full reasoning-construction layer. However, it may operate with, receive input from, or interoperate with one or more reasoning-construction components, including compositional primitives, Aurora Block processing, Persistent Existence Frame state, retrieval systems, policy engines, evidence registries, authority registries, source-trust systems, or application state.

The same admissibility invariant applies across full-stack, reduced, and hybrid embodiments: commitment is permitted only when applicable admissibility conditions are satisfied. Where admissibility conditions are not satisfied, the system produces refusal, constrained clarification, containment, escalation, delayed output, withholding, maintained unresolved state, or another governed non-commitment outcome.

In certain embodiments involving multiple candidate interpretations, commitment to one candidate interpretation is permitted only when admissibility constraints eliminate or resolve competing interpretations. Where multiple candidate interpretations remain admissible, or where no candidate interpretation is admissible, commitment is prohibited.

In certain embodiments, the runtime governance module receives, queries, polls, subscribes to, or evaluates external orientation information from institutional, regulatory, operational, or other external systems. Where a required orientation condition is stale, expired, missing, inconsistent, superseded, or not revalidated,

continued execution is treated as a candidate commitment under changed admissibility conditions rather than as continuation of a previously admitted state.

The invention further provides admissibility-state indicators and governance records. An admissibility-state indicator represents a governed epistemic or commitment-control state associated with a candidate output, state, claim, determination, action, or interpretation. Such indicator may be exposed to downstream systems or external interfaces, or may be enforced internally without external signalling. Exposure of an admissibility-state indicator does not itself constitute commitment.

A governance record or ledger may record information sufficient to reconstruct, replay, audit, verify, or evaluate an admissibility determination. Such information may include trace identifiers, state hashes, input states, candidate outputs, admissibility outcomes, failed constraints, evidence references, source references, authority references, consequence grade, refusal reason, clarification path, escalation path, collapse reason, timestamp, event ordering information, or other relevant governance metadata.

5. DEFINITIONS

For purposes of this specification:

"Commitment" means a permitted transition from a candidate, unresolved, provisional, model-originated, or internally represented state to an output, determination, claim, action, instruction, recommendation, execution, delivery, or downstream reliance condition. Commitment is not necessarily a belief update.

"Admissibility" means satisfaction of one or more structural, evidential, authority, source, freshness, scope, policy, role, domain, or consequence-governance conditions required before commitment is permitted.

"Commitment-control layer" means a module, process, or system configured to evaluate whether commitment is permitted and to enforce release, refusal, clarification, containment, escalation, delay, withholding, or maintained unresolved state.

"Reasoning-construction layer" means a module, process, or system configured to construct, maintain, transform, or evaluate structured conceptual state.

"Runtime governance module" means a deployable embodiment of the commitment-control layer that may operate independently of the full reasoning-construction layer.

"Lens" may be used as an embodiment label for the runtime governance module. The term "Lens" does not limit the invention to a product name, brand, interface, topology, or implementation.

"Model interface" means an interface to a language generation system, large language model, transformer-based model, statistical model, retrieval system, agent, application, or other output-generating component.

"Candidate output" means output proposed by a model, application, reasoning system, agent, workflow, or other process before commitment is permitted.

"Governed system instance" means any model, application, session, agent, workflow, subordinate process, language generation system, or output-originating system operating under retained commitment authority of the commitment-control layer.

"Orientation source" means an external or internal source of current operational, institutional, regulatory, authority, evidence, policy, source, counterparty, credential, jurisdictional, or world-state information used to evaluate admissibility.

"Silent decay" means a condition in which a previously valid justification, score, authority, policy, evidence item, source state, credential, licence, trade status, sanctions status, or supplier status becomes stale, expired, superseded, contradicted, or otherwise inadmissible without a direct revocation event being received by the governed system.

6. DETAILED DESCRIPTION OF THE INVENTION

6.1 Layered Architecture

The architecture comprises at least two interoperable layers:

Layer I: Reasoning-Construction Layer

Layer II: Commitment-Control Layer

The reasoning-construction layer may include one or more of:

- (a) concept kernels;
- (b) reflective semantic topology;
- (c) roles;
- (d) domains;
- (e) spans;
- (f) compositional primitives;
- (g) primitive preconditions;
- (h) primitive update logic;
- (i) primitive collapse conditions;
- (j) interpretability evaluation;
- (k) parallel interpretation maintenance;
- (l) deterministic processing cycles;
- (m) Persistent Existence Frame state;
- (n) expression-layer decoupling.

The commitment-control layer may include one or more of:

- (a) epistemic state representation;
- (b) admissibility gate;
- (c) collapse controller;
- (d) refusal module;
- (e) constrained clarification module;
- (f) containment module;
- (g) escalation module;
- (h) audit module;
- (i) admissibility-state indicator generator;
- (j) governance ledger;
- (k) policy or authority evaluator;
- (l) release controller;
- (m) orientation source evaluator.

The reasoning-construction layer and commitment-control layer may be deployed together or separately. Separation of deployment does not imply separation of invention.

6.2 Full-Stack Embodiment

In a full-stack embodiment, input is processed by the reasoning-construction layer. The reasoning-construction layer constructs or updates structured conceptual state, applies compositional primitives, maintains valid alternative interpretations, evaluates collapse conditions, and emits candidate conceptual state.

The commitment-control layer receives the emitted conceptual state and determines whether commitment is permitted.

If admissibility conditions are satisfied, the commitment-control layer permits commitment. If admissibility conditions are not satisfied, the system refuses, clarifies, contains, escalates, delays, withholds, or preserves unresolved state.

In this embodiment, the expression layer may render an admitted, refused, clarified, contained, or escalated state into natural language, symbolic form, structured data, machine-readable response, API output, audit event, or other external representation.

6.3 Reduced Runtime Governance Embodiment

In a reduced embodiment, the commitment-control layer is deployed as a runtime governance module interposed between an application and a model interface.

The runtime governance module may receive:

- (a) a candidate model output;
- (b) a candidate action;
- (c) a candidate determination;
- (d) an epistemic state proxy;
- (e) a structured state proxy;
- (f) context supplied by an application;
- (g) retrieved evidence;
- (h) policy constraints;
- (i) source metadata;
- (j) authority metadata;
- (k) freshness metadata;
- (l) domain or consequence metadata;
- (m) user or system role information;
- (n) audit or state metadata;

(o) orientation source data.

The runtime governance module evaluates whether the candidate output, action, determination, or state may proceed to delivery, commitment, consequence, execution, or downstream reliance.

If admissibility conditions are satisfied, the runtime governance module permits release or commitment. If admissibility conditions are not satisfied, it returns refusal, constrained clarification, containment, escalation, delay, withholding, maintained unresolved state, or another governed non-commitment outcome.

The runtime governance module does not require deployment of the full reasoning-construction layer. It may nevertheless receive data from, interoperate with, or be embedded within one or more reasoning-construction components.

6.3.1 Proxy State Construction in Reduced Embodiments

In reduced embodiments where structured conceptual state is not produced by a reasoning-construction layer, the runtime governance module may construct or receive an epistemic state proxy or structured state proxy for admissibility evaluation.

Such proxy state may be constructed by one or more of:

(a) parsing candidate model output to identify candidate entities, referents, bindings, claims, or determinations;

(b) extracting slots, tags, or annotations supplied by an application;

(c) receiving partial structure from a retrieval system, policy engine, or authority registry;

(d) mapping model output tokens or segments to provisional role, domain, or span representations; or

(e) receiving explicit unresolved-ambiguity, evidence, authority, or consequence metadata from an upstream component.

Admissibility evaluation is performed on the proxy state according to the same commitment rule and admissibility conditions applicable in full-stack embodiments. Proxy construction does not constitute commitment and does not delegate admissibility authority to the model interface or expression layer.

6.4 Hybrid Deployment

In hybrid embodiments, the runtime governance module receives partial structure from one or more upstream systems.

For example:

(a) an Aurora Block may produce structured conceptual state supplied to the runtime governance module;

(b) a Persistent Existence Frame may provide echo-state, potential-state, active-state, unresolved-state, or reconstruction metadata;

(c) a compositional primitive engine may provide primitive state, failed preconditions, collapse conditions, or unresolved primitive status;

(d) a retrieval system may provide evidence metadata;

(e) a policy engine may provide applicable policy constraints;

(f) an authority registry may provide authority state;

(g) a source-trust system may provide source reliability or provenance state;

(h) an application may provide consequence grade, user role, or action context;

(i) a language model may provide candidate output subject to governance.

(j) an orientation source may provide operational, regulatory, institutional, counterparty, sanctions, trade-status, credential, licence, HR, compliance, jurisdictional, or world-state information relevant to admissibility.

The runtime governance module may use any combination of such inputs to determine admissibility.

6.5 Commitment Rule

The general commitment rule is:

Commitment is permitted only when applicable admissibility conditions are satisfied.

In cases involving multiple candidate interpretations, commitment to one interpretation is permitted only when admissibility constraints eliminate or resolve competing interpretations. If more than one interpretation remains admissible, commitment is prohibited. If no interpretation is admissible, commitment is prohibited.

This rule generalizes beyond ambiguity resolution. Admissibility may fail because of unresolved referent, insufficient evidence, stale evidence, missing authority, source untrustworthiness, scope mismatch, failed primitive precondition, domain conflict, policy conflict, missing escalation path, consequence-grade mismatch, or other structural condition.

6.6 Admissibility Conditions

Admissibility conditions may include one or more of:

- (a) structural coherence;
- (b) role binding;
- (c) referent resolution;
- (d) domain compatibility;
- (e) span compatibility;
- (f) primitive precondition satisfaction;
- (g) primitive collapse condition satisfaction;
- (h) interpretability threshold satisfaction;
- (i) evidence sufficiency;
- (j) source trust;
- (k) authority validation;
- (l) freshness or staleness evaluation;
- (m) scope match;
- (n) policy compliance;
- (o) consequence grade;
- (p) escalation availability;
- (q) user role;
- (r) system role;
- (s) jurisdictional or institutional rule;
- (t) provenance;
- (u) auditability;
- (v) contradiction status;
- (w) unresolved ambiguity status;

- (x) external declared uncertainty;
- (y) persistent state consistency;
- (z) orientation source currency;
- (aa) counterparty status;
- (bb) sanctions or trade-restriction status;
- (cc) credential or licence validity.

The admissibility conditions may be deterministic, rule-based, constraint-based, or structurally evaluated. They are not limited to probabilistic confidence.

6.6A External Orientation and Silent Decay

In certain embodiments, the runtime governance module receives, queries, polls, subscribes to, or evaluates external orientation information from one or more institutional, regulatory, operational, policy, authority, source, evidence, counterparty, sanctions, trade-status, geographic-risk, credential, licence, HR, compliance, or registry systems.

Such orientation information may represent a current or previously valid world-state, institutional-state, authority-state, source-state, policy-state, or counterparty-state condition relevant to admissibility.

In some cases, the external condition may change without a direct revocation event, notification, or policy update being delivered to the governed system. The runtime governance module may therefore evaluate freshness by expiry, time-to-live, revalidation interval, source version, registry version, effective date, supersession status, jurisdictional change, or other staleness threshold.

Where a required orientation condition is stale, expired, missing, inconsistent, superseded, or not revalidated, continued execution is treated as a candidate commitment under changed admissibility conditions rather than as continuation of a previously admitted state.

The system may then refuse, contain, suspend, escalate, delay, require re-attestation, or require updated evidence or authority before permitting further commitment, delivery, execution, or downstream reliance.

A stale, missing, superseded, inconsistent, or unvalidated orientation condition does not constitute permission to continue.

6.7 Admissibility-State Indicator

In certain embodiments, the commitment-control layer produces an admissibility-state indicator.

The admissibility-state indicator may represent:

- (a) admissible;
- (b) inadmissible;
- (c) unresolved;
- (d) refused;
- (e) contained;
- (f) clarification required;
- (g) escalation required;
- (h) evidence insufficient;
- (i) authority missing;
- (j) source untrusted;
- (k) stale evidence;
- (l) scope mismatch;
- (m) unresolved referent;
- (n) failed primitive precondition;

- (o) failed policy condition;
- (p) orientation stale or expired;
- (q) other governed state.

The indicator may be exposed to a user, application, downstream system, audit system, or governance interface. Alternatively, the indicator may be enforced internally without external signalling.

Exposure of the indicator does not itself constitute commitment. The indicator represents governance state.

6.8 Governance Ledger and Audit Record

In certain embodiments, the commitment-control layer records governance events in a governance ledger or audit record.

The governance ledger may comprise append-only records, cryptographic records, structured logs, database entries, event streams, signed receipts, state hashes, or any mechanism sufficient to retain governance information.

A governance record may include:

- (a) trace identifier;
- (b) timestamp or event ordering data;
- (c) session identifier;
- (d) governed system identifier;
- (e) input state;
- (f) candidate output;
- (g) model identifier;
- (h) application identifier;

- (i) structured state proxy;
- (j) epistemic state;
- (k) admissibility-state indicator;
- (l) outcome;
- (m) failed constraints;
- (n) satisfied constraints;
- (o) refusal reason;
- (p) clarification request;
- (q) escalation path;
- (r) evidence references;
- (s) source references;
- (t) authority references;
- (u) freshness status;
- (v) consequence grade;
- (w) policy reference;
- (x) state hash;
- (y) signature or verification metadata;
- (z) release or non-release status;
- (aa) orientation source version or freshness at decision time.

The governance ledger enables reconstruction, replay, audit, verification, or evaluation of admissibility determinations.

6.9 Transport-Layer Governance

In certain embodiments, governance occurs at a transport, interface, proxy, middleware, API, gateway, orchestration, or application-boundary layer.

The runtime governance module may intercept candidate outputs before they are delivered to:

- (a) a user interface;
- (b) an application interface;
- (c) a downstream system;
- (d) an automated workflow;
- (e) a robotic system;
- (f) a medical, legal, financial, safety, or operational decision system;
- (g) another model;
- (h) an agent;
- (i) an execution engine;
- (j) a public communication channel.

The governance relationship is independent of deployment topology. It may operate locally, remotely, in-process, out-of-process, as middleware, as an API gateway, as a proxy, as an embedded library, as a service, or as a distributed control plane.

6.10 Multi-System and Multi-Session Governance

A single commitment-control layer may retain governance authority across multiple governed system instances, sessions, model interfaces, applications, or workflows.

The system may preserve unresolved epistemic states across turns, sessions, governed systems, or application contexts. New inputs may update state without forcing collapse.

The system may apply admissibility conditions consistently across originating systems, such that a candidate output cannot evade governance by being generated in a different session, model, application, or interface.

6.11 Relationship Between Full Architecture and Reduced Deployment

The reduced runtime governance module is not a separate invention from the integrated architecture. It is a deployment form of the commitment-control layer.

The full architecture may construct structured conceptual state and provide rich state to the commitment-control layer. The reduced deployment may operate on proxies, metadata, candidate outputs, or externally supplied constraints.

Both forms preserve the same invariant: commitment is governed by admissibility before consequence.

7. EXAMPLE EMBODIMENTS

7.1 Ambiguous Referent

Input:

"Emma told Lucy that her sister was arriving."

The system maintains at least two candidate interpretations:

Interpretation A: "her sister" refers to Emma's sister.

Interpretation B: "her sister" refers to Lucy's sister.

If no admissibility condition eliminates either interpretation, commitment to either referent is prohibited. The system may return a constrained clarification request identifying the missing referent.

If later context eliminates one interpretation, commitment may become admissible.

7.2 Model-Interface Governance

A language generation model produces a candidate answer that resolves an ambiguous referent. The runtime governance module receives the candidate output and relevant context. The governance module determines that competing interpretations remain admissible. The candidate output is intercepted and is not delivered. The governance module returns a refusal or constrained clarification.

7.3 Hybrid Aurora-Lens Deployment

An Aurora Block constructs structured conceptual state using roles, domains, spans, primitives, and parallel interpretations. A runtime governance module receives the structured state and determines whether commitment is admissible. The governance module permits output only if admissibility conditions are satisfied.

7.4 Healthcare Safety Example

A model generates a candidate clinical recommendation. The runtime governance module evaluates evidence state, source freshness, authority state, user role, consequence grade, and policy constraints. If required evidence or authority is missing, the recommendation is refused, contained, or escalated before delivery.

7.5 Audit Example

A candidate output is refused due to unresolved referent and insufficient authority. The governance ledger records a trace identifier, input state, candidate output, failed constraints, admissibility-state indicator, refusal reason, and state hash. The event can later be reconstructed or audited.

7.6 Procurement Freshness / Trade-Status Example

A procurement agent proposes approval of a purchase order based on a stored supplier risk score. The runtime governance module receives or evaluates supplier status, jurisdictional risk, trade restriction status, authority metadata, evidence freshness, and policy constraints from one or more orientation sources. If the stored supplier score is stale, expired, superseded, or inconsistent with a current external orientation source, the system treats further purchase-order approval as a new candidate commitment under changed admissibility conditions. The system refuses, contains, suspends, escalates, or requires re-attestation before the purchase order may proceed. A stale supplier risk score does not constitute permission to approve.

8. ADVANTAGES

The invention provides:

- (a) preservation of full architecture scope while allowing reduced deployment;
- (b) governance of candidate outputs before delivery or consequence;
- (c) compatibility with external language generation systems without delegating epistemic authority to them;
- (d) hybrid operation with structured reasoning systems, persistent state systems, retrieval systems, policy engines, and model interfaces;
- (e) explicit admissibility-state indicators;
- (f) replayable governance records;
- (g) refusal as correct governed behavior;

(h) prevention of premature commitment in ambiguity, insufficient evidence, authority failure, source failure, policy failure, or other inadmissible states;

(i) detection and governance of silent admissibility decay arising from stale, expired, superseded, or unvalidated external orientation conditions;

(j) treatment of continued execution under changed conditions as a new candidate commitment requiring fresh admissibility evaluation.

9. CLAIMS

Claim 1 — Integrated Architecture with Reduced Deployment

A computer-implemented machine reasoning governance system comprising:

(a) a reasoning-construction layer configured to construct or maintain structured conceptual state;

(b) a commitment-control layer configured to evaluate, based on structural admissibility constraints rather than probabilistic confidence alone, whether a candidate state, output, determination, inference, claim, recommendation, or action is admissible for commitment;

(c) wherein the commitment-control layer is deployable either:

(i) in full-stack operation using structured conceptual state produced by the reasoning-construction layer; or

(ii) in reduced runtime operation using a candidate output, structured state proxy, epistemic state proxy, or decision context without requiring deployment of the full reasoning-construction layer; and

(d) wherein the commitment-control layer permits commitment only when applicable admissibility conditions are satisfied, and otherwise produces refusal or constrained clarification as governed terminal outcomes, or alternatively containment, escalation, delayed output, withholding, or maintained unresolved state.

Claim 2 — Reasoning-Construction Layer

The system of claim 1, wherein the reasoning-construction layer comprises one or more of concept kernels, reflective semantic topology, roles, domains, spans, compositional primitives, interpretability evaluation, parallel interpretation maintenance, deterministic processing cycle, or Persistent Existence Frame state.

Claim 3 — Reduced Runtime Governance Module

The system of claim 1, wherein the reduced runtime operation comprises a runtime governance module interposed between an application and a model interface.

Claim 4 — Model Interface

The system of claim 3, wherein the model interface comprises a language generation model, large language model, transformer-based model, statistical model, agent, retrieval system, application module, or other output-generating system.

Claim 5 — Non-Authoritative Expression Layer

The system of claim 4, wherein the model interface operates as a non-authoritative expression layer or candidate-output generator and does not determine admissibility of commitment.

Claim 6 — Hybrid Deployment

The system of claim 3, wherein the runtime governance module receives state, constraints, metadata, orientation information, or admissibility information from one or more of a reasoning-construction layer, compositional primitive engine, Aurora Block, Persistent Existence Frame, retrieval system, source registry, authority registry, policy engine, application layer, model interface, orientation source, sanctions registry, trade-status system, supplier registry, credential registry, licence registry, HR system, compliance system, or operational-state system.

Claim 7 — Commitment Rule

The system of claim 1, wherein commitment is permitted only when applicable admissibility conditions are satisfied.

Claim 8 — Multiple Candidate Interpretations

The system of claim 7, wherein, when multiple candidate interpretations exist, commitment to one candidate interpretation is permitted only when admissibility constraints eliminate or resolve competing candidate interpretations.

Claim 9 — Prohibition of Commitment

The system of claim 8, wherein commitment is prohibited when more than one candidate interpretation remains admissible or when no candidate interpretation is admissible.

Claim 10 — Admissibility Conditions

The system of claim 1, wherein the admissibility conditions comprise one or more of structural coherence, role binding, referent resolution, domain compatibility, span compatibility, primitive precondition satisfaction, primitive collapse condition satisfaction, interpretability threshold, evidence sufficiency, source trust, authority validation, freshness, staleness, scope match, policy compliance, consequence grade,

escalation availability, user role, system role, provenance, auditability, contradiction status, unresolved ambiguity status, persistent state consistency, orientation source currency, counterparty status, sanctions status, trade-restriction status, credential validity, or licence validity.

Claim 11 — External Orientation Source

The system of claim 10, wherein one or more admissibility conditions are evaluated using orientation information received from, queried from, or associated with an external or institutional source comprising one or more of a regulatory registry, sanctions list, trade-status system, supplier registry, authority registry, credential system, HR system, policy register, evidence repository, source-trust registry, or operational state system.

Claim 12 — Silent Decay / Staleness Condition

The system of claim 11, wherein continued execution is treated as inadmissible, unresolved, or requiring revalidation when an orientation condition, evidence item, authority state, policy state, source state, supplier status, credential status, or registry status is stale, expired, superseded, inconsistent, unavailable, or not revalidated within a predetermined interval.

Claim 13 — Continued Execution as Candidate Commitment

The system of claim 12, wherein, after an admissibility-relevant change, expiry, staleness condition, or unresolved orientation condition is detected or inferred, each further execution, approval, release, recommendation, or downstream reliance is evaluated as a new candidate commitment requiring admissibility evaluation.

Claim 14 — Admissibility-State Indicator

The system of claim 1, wherein the commitment-control layer produces an admissibility-state indicator representing a governed state associated with the candidate state, output, determination, inference, claim, recommendation, or action.

Claim 15 — Indicator Exposure

The system of claim 14, wherein the admissibility-state indicator is exposed to an external interface, application, downstream system, user interface, audit interface, or governance interface.

Claim 16 — Indicator Without Commitment

The system of claim 14, wherein exposure of the admissibility-state indicator does not constitute commitment.

Claim 17 — Silent Enforcement

The system of claim 14, wherein the admissibility-state indicator is enforced internally without external signalling.

Claim 18 — Governance Ledger

The system of claim 1, further comprising a governance ledger configured to record governance events associated with candidate outputs, admissibility evaluations, refusals, clarifications, containments, escalations, releases, or maintained unresolved states.

Claim 19 — Governance Record Fields

The system of claim 18, wherein a governance record comprises one or more of a trace identifier, timestamp, event ordering data, session identifier, governed system identifier,

input state, candidate output, model identifier, application identifier, structured state proxy, epistemic state, admissibility-state indicator, governance outcome, failed constraint, satisfied constraint, refusal reason, clarification request, escalation path, evidence reference, source reference, authority reference, freshness status, consequence grade, policy reference, state hash, signature, verification metadata, orientation source version, orientation source freshness status, or release status.

Claim 20 — Audit and Replay

The system of claim 18, wherein the governance ledger enables reconstruction, replay, audit, verification, or evaluation of an admissibility determination.

Claim 21 — Transport-Layer Governance

The system of claim 3, wherein the runtime governance module is implemented as middleware, proxy, API gateway, service, embedded library, orchestration layer, application-boundary module, or distributed control plane.

Claim 22 — Pre-Consequence Interception

The system of claim 3, wherein the runtime governance module intercepts candidate output before delivery, commitment, consequence, execution, downstream reliance, or transmission to an output destination.

Claim 23 — Multi-System Governance

The system of claim 1, wherein a single commitment-control layer governs candidate outputs originating from multiple governed system instances, sessions, model interfaces, applications, agents, or workflows.

Claim 24 — Persistent Unresolved State

The system of claim 1, wherein unresolved states persist across turns, sessions, governed system instances, or application contexts until admissibility conditions permit commitment or require a governed non-commitment outcome.

Claim 25 — Persistent Existence Frame Input

The system of claim 6, wherein the runtime governance module receives active-state, echo-state, potential-state, reconstruction, projection, or unresolved-state metadata from a Persistent Existence Frame.

Claim 26 — Primitive State Input

The system of claim 6, wherein the runtime governance module receives primitive state, failed primitive preconditions, primitive collapse conditions, or unresolved primitive status from a compositional primitive engine.

Claim 27 — Aurora Block Input

The system of claim 6, wherein the runtime governance module receives structured conceptual state emitted by an Aurora Block.

Claim 28 — Claim Evaluation Protocol

The system of claim 1, wherein the commitment-control layer evaluates an external assertion by decomposing the assertion into structural elements and evaluating each structural element against maintained epistemic state or admissibility constraints.

Claim 29 — Insufficient Correspondence

The system of claim 28, wherein commitment regarding the external assertion is refused when structural correspondence is insufficient.

Claim 30 — Governed Evaluation Record

The system of claim 28, wherein the commitment-control layer, upon evaluating the external assertion, produces an admissibility-state indicator representing the governance outcome and records a governance event in the governance ledger comprising the structural elements evaluated, the maintained epistemic state or admissibility constraints evaluated against, any failed or satisfied constraints, and the admissibility outcome.

Claim 31 — Method Claim

A computer-implemented method comprising:

- (a) receiving a candidate output, candidate determination, candidate action, structured state, structured state proxy, epistemic state proxy, or decision context;
- (b) evaluating one or more admissibility conditions associated with the received candidate or state;
- (c) permitting commitment only when the admissibility conditions are satisfied;
- (d) otherwise producing refusal, constrained clarification, containment, escalation, delayed output, withholding, or maintained unresolved state; and
- (e) optionally recording a governance event associated with the evaluation.

Claim 32 — Reduced Runtime Method

A computer-implemented method performed by a runtime governance module interposed between an application and a model interface, comprising:

- (a) receiving a candidate output from the model interface;
- (b) evaluating admissibility of the candidate output before delivery or consequence;
- (c) permitting delivery only when admissibility conditions are satisfied; and
- (d) otherwise intercepting the candidate output and returning refusal, constrained clarification, containment, escalation, delayed output, withholding, or maintained unresolved state.

Claim 33 — Hybrid Method

The method of claim 32, further comprising receiving state, constraints, metadata, or admissibility information from one or more of a reasoning-construction layer, compositional primitive engine, Aurora Block, Persistent Existence Frame, retrieval system, source registry, authority registry, policy engine, application layer, or model interface.

Claim 34 — Computer-Readable Medium

A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause the one or more processors to perform the method of claim 31, 32, or 33.

10. ABSTRACT

A computer-implemented machine reasoning governance system integrates a reasoning-construction layer with a commitment-control layer and supports reduced or hybrid runtime deployment of the commitment-control layer. The reasoning-construction layer constructs structured conceptual state using concept kernels, reflective topology, compositional primitives, parallel interpretation maintenance,

deterministic processing, or Persistent Existence Frame state. The commitment-control layer evaluates whether a candidate output, determination, inference, recommendation, or action is admissible for commitment based on structural admissibility constraints rather than probabilistic confidence alone. The commitment-control layer operates in full-stack form using structured conceptual state or in reduced runtime form using candidate outputs, state proxies, or decision context without requiring the full reasoning-construction layer. Commitment is permitted only when admissibility conditions are satisfied; otherwise the system produces refusal, constrained clarification, containment, escalation, delayed output, withholding, or maintained unresolved state. In certain embodiments, the system evaluates external orientation sources and treats continued execution under stale, expired, or superseded conditions as a new candidate commitment requiring fresh admissibility evaluation. The system produces admissibility-state indicators and governance records enabling reconstruction, replay, audit, or verification of governance determinations.

END OF PROVISIONAL SPECIFICATION